

1 Solution — GIAM 5.3

Problem 6

1.1 Problem

Give an inductive proof that $\forall n \in \mathbb{N}, 5 \mid n^5 - 5n^3 + 14n$.

1.2 Answer

$$f(k+1) = f(k) + 5(k^4 + 2k^3 - k^2 - 2k + 2)$$

Here $f(n) = n^5 - 5n^3 + 14n$. The step adds a multiple of 5, so divisibility passes from k to $k+1$. Verified for every n up to 5000.

1.3 The Proof

Theorem. For every natural number n , 5 divides $f(n) = n^5 - 5n^3 + 14n$.

Proof. By induction on n .

Basis. At $n = 0$, $f(0) = 0$, and $5 \mid 0$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $f(k) = 5m$ for some integer m . Expanding with the binomial theorem,

$$\begin{aligned} f(k+1) &= (k+1)^5 - 5(k+1)^3 + 14(k+1) \\ &= f(k) + (5k^4 + 10k^3 + 10k^2 + 5k + 1) - (15k^2 + 15k + 5) + 14 \\ &= f(k) + 5(k^4 + 2k^3 - k^2 - 2k + 2). \end{aligned}$$

Substituting the hypothesis,

$$f(k+1) = 5(m + k^4 + 2k^3 - k^2 - 2k + 2),$$

and the bracket is an integer, so $5 \mid f(k+1)$.

Basis and step together give the result for every n . ■

As in 5.3 Problem 3, every binomial coefficient $\binom{5}{j}$ with $0 < j < 5$ is a multiple of 5. That is why the $k^4, \dots, k$ terms of $(k+1)^5$ fall into line, and the leftover constants $1 - 5 + 14 = 10$ happen to be a multiple of 5 too.

1.4 The Picture

The polynomial hides a product of five consecutive integers, drawn below.

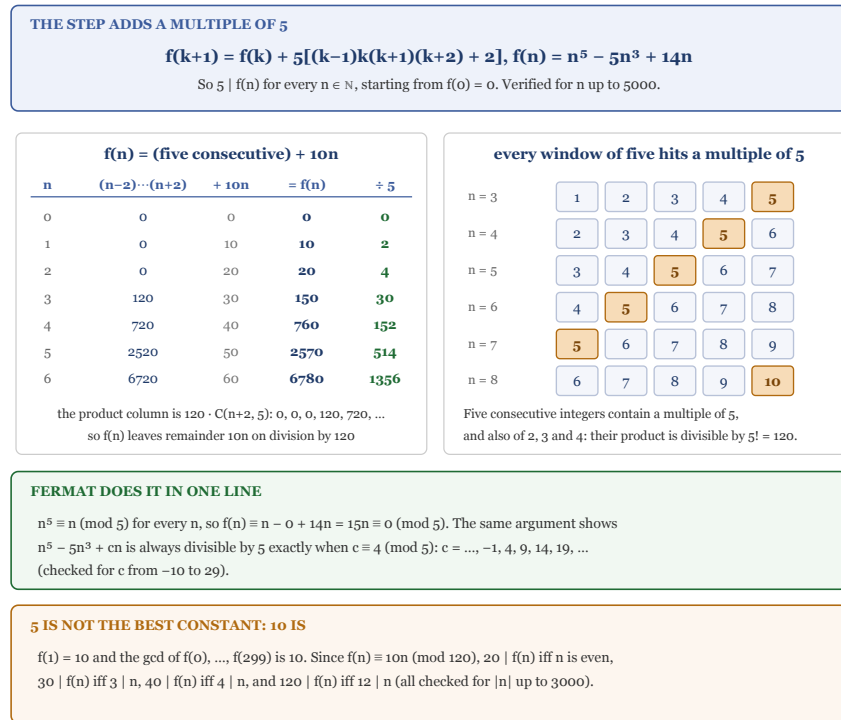


Figure 1.1: A diagram in four parts. The top band states the step identity. The left panel tabulates, for n from zero to six, the product of the five consecutive integers n minus two through n plus two (0, 0, 0, 120, 720, 2520, 6720), plus ten n , giving f of n and f of n over five. The right panel shows windows of five consecutive integers for n from three to eight, each containing a highlighted multiple of five. A green band gives the one-line Fermat proof, and an orange band notes that ten is the best constant, with 20, 30, 40 and 120 dividing f of n exactly when n is a multiple of 2, 3, 4 and 12.

1.5 Remark — Fermat's Little Theorem Does It in One Line

In 5.3 Problem 3 we proved $11 \mid x^{11} + 10x$, which says $x^{11} \equiv x \pmod{11}$, by exactly this kind of induction. The same argument with 5 in place of 11 gives $n^5 \equiv n \pmod{5}$ for every n , and then, working modulo 5,

$$f(n) = n^5 - 5n^3 + 14n \equiv n - 0 + 14n = 15n \equiv 0 \pmod{5}.$$

This also shows what made the coefficient 14 work. The polynomial $n^5 - 5n^3 + cn$ is divisible by 5 for every n exactly when $1 + c \equiv 0$, that is, $c \equiv 4 \pmod{5}$. A search over c from -10 to 29 found exactly $-6, -1, 4, 9, 14, 19, 24, 29$. The middle term $-5n^3$ plays no part modulo 5.

1.6 Remark — Five Consecutive Integers in Disguise

The quartic in the step factors neatly:

$$k^4 + 2k^3 - k^2 - 2k = (k-1)k(k+1)(k+2).$$

That is a product of four consecutive integers. It suggests looking at f itself the same way, and indeed

$$f(n) = (n-2)(n-1)n(n+1)(n+2) + 10n,$$

verified for $|n|$ up to 3000. Among any five consecutive integers one is a multiple of 5, one of 4, one of 3, and another is even, and in fact their product is $120 \binom{n+2}{5}$, a multiple of $5! = 120$. So $f(n)$ is a multiple of 120 plus $10n$, which gives a direct proof without induction.

The step identity becomes a small instance of a familiar pattern. Writing $P(n)$ for the product of five consecutive integers centred at n ,

$$P(k+1) - P(k) = 5(k-1)k(k+1)(k+2),$$

the discrete counterpart of $\frac{d}{dx}x^5 = 5x^4$. Dividing by 120, it is Pascal's rule $\binom{k+3}{5} - \binom{k+2}{5} = \binom{k+2}{4}$, the identity behind the hockey-stick sums of 5.2. Adding the 10 from the $10n$ term gives exactly the boxed increment.

1.7 Remark — Is 5 the Best Divisor?

No. Since 120 divides $P(n)$, $f(n) \equiv 10n \pmod{120}$, so 10 divides every value. That is sharp, because $f(1) = 10$, and the gcd of $f(0), \dots, f(299)$ is 10.

The congruence also settles every divisor of 120 at once: $d \mid f(n)$ exactly when $d \mid 10n$. In particular

$$\begin{aligned} 20 \mid f(n) &\iff 2 \mid n, & 30 \mid f(n) &\iff 3 \mid n, \\ 40 \mid f(n) &\iff 4 \mid n, & 120 \mid f(n) &\iff 12 \mid n, \end{aligned}$$

all checked for $|n|$ up to 3000. This is the pattern of 5.3 Problems 1 and 2, where the stated divisor understated the truth, rather than Problems 3 and 5, where it was sharp. Since f is an odd function, $f(-n) = -f(n)$, everything here holds for negative integers as well.